

On a Remarkable Polyhedron with One-Sided Total Surface

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The purpose of the following pages is to pursue, for polyhedra and especially for one particular polyhedron, several facts concerning one-sided surfaces which are known from analysis situs.

1. Derivation and description of the one-sided heptahedron.

We begin with the consideration of a regular octahedron. Let its center be O , and let its vertices, grouped into three pairs of opposite vertices, be $X_-, X_+; Y_-, Y_+; Z_-, Z_+$. For a more definite picture, suppose the octahedron placed as in Fig. 1, so that the diagonal X_-X_+ is directed from left to right, Y_-Y_+ from front to back, and Z_-Z_+ from below upward. We divide the eight octahedral faces into two groups, the positive and the negative, assigning to each face the sign of the product of the signs used in denoting its vertices. Thus the positive faces are

$$\alpha = X_+Y_-Z_-, \quad \beta = X_-Y_+Z_-, \quad \gamma = X_-Y_-Z_+, \quad \delta = X_+Y_+Z_+,$$

and the negative faces are

$$\alpha' = X_-Y_+Z_+, \quad \beta' = X_+Y_-Z_+, \quad \gamma' = X_+Y_+Z_-, \quad \delta' = X_-Y_-Z_-.$$

Beside these two groups of faces we consider a third, the group of squares

$$\xi = Y_-Z_-Y_+Z_+, \quad \eta = Z_-X_-Z_+X_+, \quad \zeta = X_-Y_-X_+Y_+.$$

Through each edge of the octahedron there passes one face from each of these three groups. Hence, if we omit any one of the three groups, the remaining faces form a closed polyhedron, since two of them pass through each edge. Thus, in addition to the octahedron, by combining the squares with the positive or the negative octahedral faces respectively, we obtain two mutually similar heptahedra. All three polyhedra have the same vertices and edges. The octahedron can be brought onto itself in 24 ways by rotations about O ; in addition, there are 24 symmetric operations, that is, reflections and rotatory reflections, which do the same. Of these operations, one half – namely 12 rotations and 12 symmetric operations – carry each heptahedron into itself, while the other half interchange the two heptahedra.

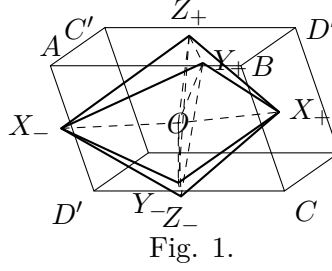
In what follows we refer to the heptahedron formed from the faces

$$\begin{aligned} \alpha &= X_+Y_-Z_-, \quad \beta = X_-Y_+Z_-, \quad \gamma = X_-Y_-Z_+, \quad \delta = X_+Y_+Z_+, \\ \xi &= Y_-Z_-Y_+Z_+, \quad \eta = Z_-X_-Z_+X_+, \quad \zeta = X_-Y_-X_+Y_+ \end{aligned} \tag{1}$$

and call it, for short, the “one-sided heptahedron”. It is indeed one-sided. Suppose we move on the side of the face $X_+Y_+Z_+$ away from O (Fig. 1). Crossing the edge X_+Y_+ , we reach the lower side of $X_+Y_+X_-Y_-$; then, crossing X_-Y_- , the side of $X_-Y_-Z_+$ away from O ; then, crossing Y_-Z_+ , the right side of $Y_-Z_-Y_+Z_+$; and finally, crossing Y_+Z_+ , the side of the original face $X_+Y_+Z_+$ facing the center.

Without appealing to the picture, one can prove the one-sidedness of the polyhedron in the following way, after Möbius. Consider the sequence of traversed polygons, each adjacent to the next, while the last coincides with the first. Give the first an arbitrary direction of circulation, say $X_+Y_+Z_+$, and determine the direction of circulation of each following polygon so that the common edge with the preceding polygon is traversed in the opposite sense. We then obtain

$$X_+Y_+Z_+, \quad Y_+X_+Y_-X_-, \quad X_-Y_-Z_+, \quad Z_+Y_-Z_-Y_+, \quad Z_+Y_+X_+.$$

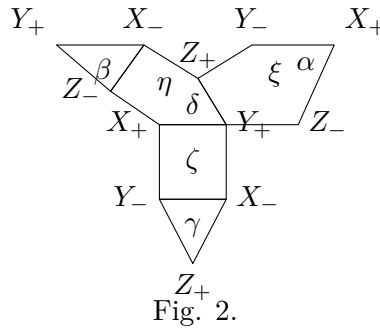


We see that the face appearing as the initial and final polygon receives opposite directions of circulation in the two cases. This shows that the path just followed leads continuously from one side of the initial face to the other. W. Dyck has pointed out, however, that the property just used – that along certain closed paths the indicatrix, or direction of circulation, is reversed – is intrinsic to the surface, while one-sidedness expresses a relation of the surface to its surrounding space and can be lost when the surface is considered in a different surrounding space, that is, in a space with different connectivity relations. In Euclidean three-space, and also in projective three-space, the one-sided surfaces are indeed precisely those with reversible indicatrix, and conversely; but in certain spaces of other kinds this is no longer so. A general theorem on the relations between one-sidedness and two-sidedness on the one hand, and reversibility and non-reversibility of the indicatrix on the other, for configurations of arbitrarily many dimensions, will be given in a second article.

If the one-sided heptahedron is cut along the edge path

$$X_+Y_-Z_+X_-Y_+Z_-X_+,$$

the surface becomes simply connected and can be unfolded into a plane as shown in Fig. 2.



The fact that this transformation is effected by a single closed cut shows, as analysis situs teaches, that the heptahedron, as a one-sided surface of least connectivity, is equivalent to the projective plane, that is, it can be mapped onto it one-to-one and continuously in the sense of projective geometry. This mapping will concern us below. In each of the diagonals X_-X_+ , Y_-Y_+ , Z_-Z_+ two squares intersect. The points of these diagonals are therefore double points of the total surface, except for the vertices, which are simple, and for the point O , which counts triple. In the mapping under discussion, a double point will correspond to two different points of the plane, and O to three.

2. Division of the plane by a complete quadrilateral.

The plane of projective geometry is divided by four lines a, b, c, d , no three of which pass through a point, into seven regions – four triangles and three quadrilaterals – convex in the sense of this

geometry, provided that connectivity at infinity is taken into account. Let the vertices of the complete quadrilateral be

$$\begin{aligned} (a, d) &= X_-, & (b, c) &= X_+, \\ (b, d) &= Y_-, & (c, a) &= Y_+, \\ (c, d) &= Z_-, & (a, b) &= Z_+. \end{aligned} \tag{2}$$

On each of the lines a, b, c, d there are three vertices; by the segment (edge) determined by two of them we always mean, of course, the segment, finite or infinitely long, which does not contain the third vertex. With this convention, the seven polygons are determined by the cyclic order of their vertices. We obtain the triangles

$$\alpha = X_+Y_-Z_-, \quad \beta = X_-Y_+Z_-, \quad \gamma = X_-Y_-Z_+, \quad \delta = X_+Y_+Z_+,$$

and the quadrilaterals

$$\xi = Y_-Z_-Y_+Z_+, \quad \eta = Z_-X_-Z_+X_+, \quad \zeta = X_-Y_-X_+Y_+$$

(Fig. 3). This notation agrees exactly with the scheme (1) set up for the heptahedron. The elements – vertices, edges, and faces – of the heptahedron are therefore grouped in the same way as in the division of the plane by the complete quadrilateral; in Eberhard’s terminology, the two configurations are “isomorphic”.

It is now easy to see the possibility of mapping the heptahedron and the projective plane point by point, one-to-one and continuously in the sense of projective geometry, so that each heptahedron face corresponds to the polygon of the plane division bearing the same name. To construct such a mapping, one may first put equally named edges continuously into correspondence and then, in infinitely many ways, establish for each pair of equally named faces a one-to-one continuous mapping which respects the already fixed correspondence on the edges.

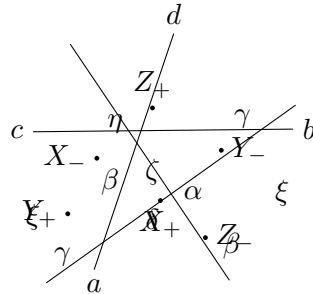


Fig. 3.

3. The general one-sided heptahedron.

Every polyhedron is transformed by any collineation into an isomorphic one. If we are to admit arbitrary collineations, we must of course also admit polyhedra extending to infinity. Only for a few polyhedra with small number of faces does the converse of this statement hold, namely that isomorphic polyhedra must also be collinear. For the one-sided heptahedron it does hold, provided we impose only the conditions that *the polyhedron is genuinely a spatial figure and that its faces are convex*.¹ If one tries to construct, in the most general way, a polyhedron with the seven faces

$$\begin{aligned} &X_+Y_+Z_+, X_+Y_-Z_-, X_-Y_+Z_-, X_-Y_-Z_+, \\ &Y_+Z_+Y_-Z_-, Z_+X_+Z_-X_-, X_+Y_+X_-Y_-, \end{aligned}$$

¹We keep this assumption throughout.

one sees that, since of the three lines X_-X_+ , Y_-Y_+ , Z_-Z_+ each pair is to lie in a quadrilateral plane and hence to intersect, the three lines must either all pass through a point or all lie in a plane. The latter is excluded by the requirement that the figure be spatial. The three lines therefore have a common point O . This point must not coincide with any of the six vertices, if genuine quadrilaterals are to arise. Thus, to construct the heptahedron, one first draws through a point O three lines not lying in one plane and chooses on each of them two distinct points different from O , otherwise arbitrary, $X_-, X_+; Y_-, Y_+; Z_-, Z_+$. Once this has been done, the rest of the construction can be carried out always and in only one way.

At the construction of each edge, for example X_-Y_+ , there is at first an ambiguity as to which of the two joining segments completing the whole line is to be taken. This ambiguity is removed by the condition of convexity. For example, if $X_-Y_-X_+Y_+$ is to be a convex quadrilateral, then the intersection of the unlimited lines Y_+X_- and Y_-X_+ must lie outside the bounded sides Y_+X_- and Y_-X_+ . Accordingly, the six opposite pairs of edges of the heptahedron

$$\begin{array}{lll} Y_+Z_+, & Y_-Z_-, & Z_+X_+, \\ Z_-X_-, & X_+Y_+, & X_-Y_-, \\ Y_+Z_-, & Y_-Z_+, & Z_+X_-, \\ Z_-X_+, & X_+Y_-, & X_-Y_+ \end{array}$$

are to be constructed so that only the extensions of the edges of each pair intersect. These edges then bound not only the three convex quadrilaterals

$$Y_-Z_-Y_+Z_+, \quad Z_-X_-Z_+X_+, \quad X_-Y_-X_+Y_+,$$

but also the four triangles

$$X_+Y_-Z_-, \quad X_-Y_+Z_-, \quad X_-Y_-Z_+, \quad X_+Y_+Z_+.$$

To show, for instance, that the edges Y_+Z_+ , Z_+X_+ , X_+Y_+ bound a triangle and not an odd open polygonal path, it suffices to exhibit a line or plane which meets not these segments themselves but their complementary segments. By our construction this is the case for the plane $X_-Y_-Z_-$, which contains the opposite edges Y_+Z_+ , Z_+X_+ , X_+Y_+ .

Given two such heptahedra, with their vertices denoted by the same letters $X_-, X_+; Y_-, Y_+; Z_-, Z_+$, we must show that there is one and only one collineation which carries the first heptahedron onto the second, taking each vertex to the equally named one. Such a collineation must carry the planes $X_+Y_+Z_+$, $X_+Y_-Z_-$, $X_+Y_+X_-Y_-$, $Y_+Z_+Y_-Z_-$, $Z_+X_+Z_-X_-$ of the first figure to the equally named planes of the second. Since no four of these planes pass through one point, this condition determines a collineation uniquely. As is immediately seen, this collineation takes the vertices of the first heptahedron to the equally named vertices of the second; and, since convex polygons can go only to convex polygons and, as we saw, the heptahedron is determined by its vertices under the imposed convexity condition, it also takes the faces and edges of the first heptahedron to the equally named faces and edges of the second.

4. Facewise collinear mapping of the one-sided heptahedron onto the plane division determined by a complete quadrilateral.

Suppose two isomorphic polyhedra are mapped one-to-one and continuously onto each other in such a way that every straight segment on a face of one corresponds to a straight segment on the other; equivalently, each pair of corresponding faces is collinear. Such a mapping will be called “facewise collinear”. It should be expressly emphasized, however, that one-to-one-ness and continuity of the mapping are required also along the edges. Given a cube and any hexhedron

isomorphic to it and bounded by convex quadrilaterals, there is, for each pair of corresponding faces, a definite collineation carrying one to the other. But the cube edges, belonging to two faces each, then acquire two mappings, and these are in general not identical. Thus the mapping considered in this example is not facewise collinear in our sense.

A spatial collineation carrying one polyhedron to a second at the same time gives a facewise collinear mapping of the two polyhedra onto each other. But it is evident that not every such mapping must conversely be contained in a spatial collineation. In the case of the one-sided heptahedron, however, the mapping furnished by the spatial collineation is the only facewise collinear one. For first the collineations of the quadrilaterals are completely determined by the assignment of the vertices, and the projective correspondences of all the edges thereby fixed in turn determine the collineations in the triangles.

If the phrase “facewise collinear mapping” is applied, analogously, not only to polyhedra but also to divisions of the plane into regions, then a collineation of the whole plane at once supplies a facewise collinear mapping of every region-division contained in it; conversely, however, not every such mapping of an arbitrary region-division need arise from a collineation of the whole plane. But if the division is specifically one of two planes ε and ε' by complete quadrilaterals $abcd$ and $a'b'c'd'$, then exactly as for the heptahedra one proves that the collineation of the whole planes determined by assigning a, b, c, d to a', b', c', d' is the unique facewise collinear mapping of the two divisions.

The preceding discussion naturally raises the question whether, among the infinitely many possible one-to-one continuous mappings of a one-sided heptahedron onto the division of a plane ε by an isomorphically corresponding quadrilateral $abcd$, there is also a facewise collinear one. It is clear that there can be no more than one; hence it remains only to prove that one exists.

As before, denote the vertices of the heptahedron and the corresponding vertices of the quadrilateral in ε by the same letters X, Y, Z . But now $\alpha, \beta, \gamma, \delta; \xi, \eta, \zeta; \alpha', \beta', \gamma', \delta'$ denote not the bounded heptahedron faces or negative octahedron faces but the unbounded planes in which those faces lie. Extend opposite edges to meet, and denote their intersection points as follows:

$$\begin{cases} (Y_+Z_+, Y_-Z_-) = X_-^\infty, & (Y_+Z_-, Y_-Z_+) = X_+^\infty, \\ (Z_+X_+, Z_-X_-) = Y_-^\infty, & (Z_+X_-, Z_-X_+) = Y_+^\infty, \\ (X_+Y_+, X_-Y_-) = Z_-^\infty, & (X_+Y_-, X_-Y_+) = Z_+^\infty. \end{cases} \quad (3)$$

Thus on every edge we have three points, denoted by the letters X, Y, Z and signs whose product is negative. The six intersection points lie, three at a time, on four lines

$$X_-^\infty Y_+^\infty Z_+^\infty, \quad X_+^\infty Y_-^\infty Z_+^\infty, \quad X_+^\infty Y_+^\infty Z_-^\infty, \quad X_-^\infty Y_-^\infty Z_-^\infty.$$

They are therefore the vertices of a complete quadrilateral, whose plane we denote by ε^∞ ; in the heptahedron described in §1 this is the plane at infinity. As in ε^∞ , so also in each heptahedron plane there is a complete quadrilateral. They are assembled in the following table:

$$\begin{cases} (\alpha) & X_-^\infty Y_+^\infty Z_+^\infty, & X_+Y_-Z_+^\infty, & X_+Y_+^\infty Z_-, & X_-^\infty Y_-Z_-, \\ (\beta) & X_-Y_+Z_+^\infty, & X_+^\infty Y_-^\infty Z_+, & X_+^\infty Y_+Z_-, & X_-Y_-^\infty Z_-, \\ (\gamma) & X_-Y_+^\infty Z_+, & X_+^\infty Y_-Z_+, & X_+^\infty Y_+^\infty Z_-^\infty, & X_-Y_-Z_-^\infty, \\ (\delta) & X_-^\infty Y_+Z_+, & X_+Y_-^\infty Z_+, & X_+Y_+Z_-^\infty, & X_-^\infty Y_-^\infty Z_-^\infty, \\ (\xi) & X_-^\infty Y_+Z_+, & X_+^\infty Y_-Z_+, & X_+^\infty Y_+Z_-, & X_-^\infty Y_-Z_-, \\ (\eta) & X_-Y_+^\infty Z_+, & X_+Y_-^\infty Z_+, & X_+Y_+^\infty Z_-, & X_-Y_-^\infty Z_-, \\ (\zeta) & X_-Y_+Z_+^\infty, & X_+Y_-Z_+^\infty, & X_+Y_+Z_-^\infty, & X_-Y_-Z_-^\infty. \end{cases} \quad (4)$$

Each of these planes is divided by the complete quadrilateral into seven regions, one of which is the bounded heptahedron face. We now map the seven heptahedron planes collinearly onto the

plane ε by making the lines of the quadrilaterals lying in those planes correspond to the lines

$$a = X_-Y_+Z_+, \quad b = X_+Y_-Z_+, \quad c = X_+Y_+Z_-, \quad d = X_-Y_-Z_-$$

of the plane ε . Under this mapping the seven bounded heptahedron faces are collinearly mapped to the seven equally named regions of ε . To show that the mapping is also facewise collinear in the sense specified above, it remains only to verify that uniqueness on the edges is not violated. This is clear if the two projective mappings of an edge are followed not merely along the edge itself but also along its extension; for then on the line we have three points, the vertices of the complete quadrilateral, for which we already know that the two mappings assign the same images.

The complete quadrilaterals given in (4) are the intersections of the heptahedron planes with the negative octahedron planes $\alpha', \beta', \gamma', \delta'$. If the heptahedron has the semiregular form from which we started, then these planes form a regular tetrahedron whose vertices A', B', C', D' (Fig. 1), together with the vertices of the regular tetrahedron $ABCD$ formed by $\alpha, \beta, \gamma, \delta$, are the vertices of the cube circumscribed about the octahedron. The heptahedron planes are collinearly related to the plane ε and hence to each other. Moreover, as is easily seen, every triangular plane is in perspective position with every quadrilateral plane: the intersection line of such a pair of planes lies in one of $\alpha', \beta', \gamma', \delta'$, and the collineation is exactly the perspectivity whose center is the intersection point of the remaining three. In this simplest way, given a point in any one heptahedron plane, one can find the corresponding points in the others, and, given two arbitrary points on the total surface of the heptahedron, one can construct the polygonal path joining them which becomes a straight line under the mapping to the plane.

If ε is made to coincide with ε^∞ , and the points X_-X_+, Y_-Y_+, Z_-Z_+ lying in ε with $X_-^\infty X_+^\infty, Y_-^\infty Y_+^\infty, Z_-^\infty Z_+^\infty$, then we have eight collinear planes, ordered in two groups

$$\alpha, \beta, \gamma, \delta \quad \text{and} \quad \xi, \eta, \zeta, \varepsilon^\infty.$$

Every plane of one group is in perspective with every plane of the other, and the centers of perspectivity are the vertices of the tetrahedron $\alpha'\beta'\gamma'\delta'$. We are thus led to the much-studied system of three tetrahedra in desmic position: $\alpha\beta\gamma\delta$, $\xi\eta\zeta\varepsilon^\infty$, and $\alpha'\beta'\gamma'\delta'$ form such a system. In every desmic tetrahedron system there are 24 one-sided heptahedra; the planes of one of them are obtained by omitting, among the twelve planes of the desmic system, the planes of one tetrahedron and one further arbitrary plane. The conjugate desmic system likewise supplies 24 one-sided heptahedra. Thus these heptahedra occur in groups of 48, in such a way that any one of them determines the entire group.

We shall not pursue these relations further. Instead, let us add one point to the proof given above for the possibility of the facewise collinear mapping of heptahedron and plane. If the initial task is merely to give a polygonal division of the plane isomorphic to the one-sided heptahedron, the division produced by a complete quadrilateral is clearly not the only possible one. For example, one may continuously move the vertices of a complete quadrilateral and with them its twelve edges, so that three edges which previously together made up a line now form an odd closed polygonal path. As long as the displacements remain within certain bounds, the new figure (Fig. 4) will still give a polygonal division of the plane isomorphic to the heptahedron.

It is noteworthy, however, that *a facewise collinear mapping of the heptahedron onto the plane is possible only in such a way that the seven plane regions corresponding to the heptahedron faces are bounded by the lines of a complete quadrilateral.*

To prove this, formulate the mapping problem as follows. The faces of a one-sided heptahedron are to be mapped onto a plane in such a way that: 1. the mapping of each individual face is collinear; 2. the points of each edge have the same images under the two collineations belonging to the faces containing that edge; 3. the mapping of the whole heptahedron covers the plane once and without gaps.

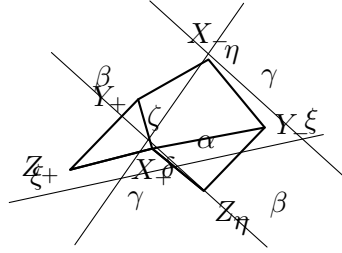


Fig. 4.

For the moment we ignore the third requirement and suppose that a mapping of the heptahedron onto a plane ε is given which satisfies 1 and 2. We denote the heptahedron vertices and faces as before. Each vertex participates in four collineations; by 2, the four images must coincide. We denote the images as the corresponding vertices. In the plane ε we also extend the opposite sides of the quadrilaterals $Y_+Z_+Y_-Z_-$, $Z_+X_+Z_-X_-$, $X_+Y_+X_-Y_-$ to meet and introduce for their intersection points the notation given in (3). We further suppose the collinear mappings extended beyond the bounded heptahedron faces to the whole planes. Then condition 2 holds also on the extensions of the edges. Looking at the collineations of the quadrilateral planes, we find that the points $X_-^\infty X_+^\infty Y_-^\infty Y_+^\infty Z_-^\infty Z_+^\infty$ lying in them correspond to the equally named points of ε . By 2 the same points, counted as belonging to the triangular planes, receive the same images under the collineations of those planes. But in the triangular planes $\alpha, \beta, \gamma, \delta$ these points lie three at a time on the lines

$$X_-^\infty Y_+^\infty Z_+^\infty, \quad X_+^\infty Y_-^\infty Z_+^\infty, \quad X_+^\infty Y_+^\infty Z_-^\infty, \quad X_-^\infty Y_-^\infty Z_-^\infty;$$

the corresponding points of ε must therefore have the same incidence.

It remains to ask under what conditions, in the plane ε , the points $X_-^\infty X_+^\infty Y_-^\infty Y_+^\infty Z_-^\infty Z_+^\infty$ derived from X_-X_+, Y_-Y_+, Z_-Z_+ have this position. This is first of all the case when X_-X_+, Y_-Y_+, Z_-Z_+ are the vertices of a complete quadrilateral; then these points coincide with $X_-^\infty X_+^\infty Y_-^\infty Y_+^\infty Z_-^\infty Z_+^\infty$. Excluding this already treated case, not every one of the four triples

$$X_-Y_+Z_+, \quad X_+Y_-Z_+, \quad X_+Y_+Z_-, \quad X_-Y_-Z_-$$

can be collinear. Suppose that $X_-Y_+Z_+$ are not collinear, and consider the two triangles

$$X_-Y_+Z_+ \quad \text{and} \quad X_+Y_-Z_-.$$

The non-collinearity of X_+, Y_-, Z_- follows from the collinear mapping of the heptahedron triangle $X_+Y_-Z_-$. Here the intersections of homologous sides

$$X = (Y_+Z_+, Y_-Z_-), \quad Y = (Z_+X_-, Z_-X_+), \quad Z = (X_-Y_+, X_+Y_-)$$

lie on a line. The points $X^\infty, Y^\infty, Z^\infty$ are distinct from X_-X_+, Y_-Y_+, Z_-Z_+ . For example, since the quadrilateral $X_+Y_+X_-Y_-$, like the collinear one of the heptahedron, must be convex, the point Z^∞ can coincide with none of X_+, Y_+, X_-, Y_- ; and since Z_+ is not on X_+Y_+ and Z_- is not on X_-Y_- , it cannot coincide with Z_+ or Z_- either. The points $X^\infty, Y^\infty, Z^\infty$ are also mutually distinct, by the collinearity relation just as in the heptahedron. The six lines $Y_+Z_+, Z_+X_-, X_-Y_+, Y_-Z_-, Z_-X_+, X_+Y_-$ are distinct; for example, neither of the two points X_-, Y_+ can lie on the line X_+Y_- . These six lines are also distinct from $X^\infty Y^\infty Z^\infty$: if, for instance, $X_+Y_- = X^\infty Y^\infty Z^\infty$, then this line would also have to contain Z_+ , the point lying on Y_-X^∞ , whereas X_+, Y_-, Z_+ cannot be collinear.

Under these circumstances, Desargues's theorem applied to the two triangles $X_+Y_+Z_+$ and $X_-Y_-Z_-$ implies that the joining lines of homologous vertices X_-X_+, Y_-Y_+, Z_-Z_+ must pass

through one point O . The resulting figure can plainly be regarded as the central projection of a spatial one-sided heptahedron onto the plane. It is then immediate that the mapping supplied by such a projection does satisfy conditions 1 and 2, but it is also easy to see that it fails to satisfy condition 3. Indeed, if a closed polyhedron lies wholly in the finite part of space and is cut by a plane, the section consists of closed finite polygonal paths, which are met by any line through no vertex in an even number of points. Such a polyhedron is therefore met by every line cutting no edge in an even number of points. The same holds for every closed polyhedron collinear with one lying wholly in the finite part of space; consequently, the projection of such a polyhedron onto a plane can cover that plane everywhere – apart from individual lines – only an even number of times, and hence not once. This reasoning applies to the one-sided heptahedron. Thus it is proved that the unique facewise collinear mapping of the one-sided heptahedron into the projective plane can yield only the region-division of a complete quadrilateral.

5. The double surface of the one-sided heptahedron.

To every one-sided surface there corresponds a definite double surface. Imagine first cutting out a simply connected piece of the one-sided surface, placing a congruent piece on one side of it, and then continuing this covering process continuously. In the end it forms a connected surface which covers the original surface everywhere on both sides. This new surface is two-sided, since on it one can distinguish the side with which it lies on the original surface from the other side. In the case of a one-sided polyhedron the new surface is again a polyhedron, with twice the number of vertices, edges, and

faces. This polyhedron also has the peculiarity that its points coincide in pairs. But this circumstance is irrelevant if one has only the type of the polyhedron in view, that is, if one does not distinguish between isomorphic polyhedra; for among the many isomorphic polyhedra there will in general certainly be some which do not have that peculiarity.

To every one-sided polyhedron type there corresponds a definite two-sided one. In the case of the one-sided heptahedron the double surface must be composed of six quadrilaterals and eight triangles. One easily recognizes in it a type which can be represented by a very familiar semiregular polyhedron. It is the *cuboctahedron*, which one can obtain, for instance, from the cube by cutting off its corners by planes as far as the midpoints of the edges issuing from them. In Figures 1 and 5 the heptahedron and cuboctahedron are placed side by side. Of the vertices of the latter, two at a time bear the same designation, namely the same designation as the isomorphically corresponding vertex of the heptahedron. From this one may verify, by reading off the faces, that the cuboctahedron is indeed isomorphic to the double surface of the heptahedron.

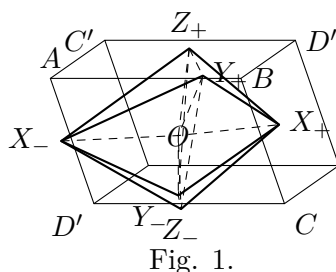


Fig. 1.

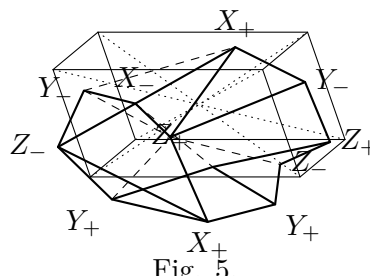


Fig. 5.

Moreover the heptahedron is taken in its semiregular form and its edge equal to that of the cuboctahedron. This gives rise to the quite special property that corresponding faces become congruent. *One can therefore cut open the cuboctahedron along certain edges, fold it together in a definite way, and finally attach together again the same points that had previously been separated by the cut, thereby transforming it into the double surface of a heptahedron without*

altering the shape or size of the individual faces. In this transformation, each two points of the cuboctahedron which lie on the same diameter drawn through its center M are brought together into the same point of the heptahedron. Thus there arises a two-to-one and facewise congruent mapping of cuboctahedron and heptahedron. The two cuboctahedron points that correspond to one heptahedron point in turn yield one and the same image if the cuboctahedron is projected from its center M onto a plane ε . From the two mappings there results a one-to-one correspondence between heptahedron and plane ε which, being composed of a facewise congruent and a facewise perspectival mapping, must be facewise collinear.

Furthermore one sees that the edges of the cuboctahedron can be arranged into four regular hexagons

$$X_-Y_+Z_+X_-Y_+Z_+, \quad X_+Y_-Z_+X_+Y_-Z_+, \quad X_+Y_+Z_-X_+Y_+Z_-, \quad X_-Y_-Z_-X_-Y_-Z_-$$

whose planes pass through M ; hence the present mapping of the cuboctahedron onto the plane ε supplies a division into regions effected by four straight lines. This again proves the possibility of mapping such a region-division facewise collinearly onto the one-sided heptahedron.

The case just treated is not the only one in which a convex cuboctahedron – by which we now mean any polyhedron isomorphic to the polyhedron otherwise so called – and the double surface of a one-sided heptahedron are facewise congruent. If in Figures 1 and 5, instead of the two congruent cubes, we imagine congruent parallelepipeds with all edges equal, and therefore with rhombic faces, and if, as before, we understand X_-X_+, Y_-Y_+, Z_-Z_+ in Fig. 1 as face midpoints and in Fig. 5 as edge midpoints, then these figures will still represent a heptahedron and a cuboctahedron with congruent corresponding faces. For each two corresponding edges are equal, namely half as large as the face-diagonal of the parallelepiped parallel to them, and the quadrilaterals are rectangles. Starting from the heptahedron, one sees that the double surface of an arbitrary rectangular heptahedron, that is, one that has arisen from an octahedron with a center and with equally long diagonals cutting one another at arbitrary angles, admits transformation into a cuboctahedron. If one restricts oneself to those cuboctahedra whose vertices are edge-midpoints of a parallelepiped, then all cases in which the transformation in question is possible have thereby been dealt with. For, even if one assumes only an ordinary parallelepiped, consideration of the parallelogram formed by the midpoints of four parallel edges – and this parallelogram is congruent to two parallelepiped faces – shows that one has a rhombus, since in the transformation to the double heptahedron all four sides come to coincide with one another. If, however, that restriction is removed, there are still other cases.

To settle the question quite generally, we first note that *every convex polyhedron which is facewise congruent to the double surface of a one-sided polyhedron must necessarily have a center.* Namely, if we map the double surface onto itself by assigning to every point the point coinciding with it, this mapping is a facewise symmetrically congruent one. Hence the convex polyhedron arising from the double surface is also facewise symmetrically congruent to itself. On the other hand, by Cauchy's polyhedron theorem, two isomorphic facewise congruent, or symmetrically congruent, polyhedra are congruent, or symmetrically congruent, as wholes. Therefore the convex polyhedron under consideration must be transformable into itself by a symmetric operation. This symmetric operation, moreover, interchanges the polyhedron points in pairs, and is therefore involutory. There are, however, only two involutory symmetric operations: reflection in a plane and reflection in a point (inversion). If reflection in a plane occurred, the polyhedron would have to lie on both sides of the reflecting plane and thus be cut by it. But then there would be points on the polyhedron which coincide with their corresponding points. That would be possible only if the polyhedron cut itself, and this is excluded by convexity. Thus only reflection in a point remains, and this point is then the center.

Suppose now that there is a convex cuboctahedron which can be transformed into the double surface of a heptahedron. Let M denote its center, and otherwise retain the earlier notation.

It then contains two quadrilaterals $\xi = Y_+Z_+Y_-Z_-$, two quadrilaterals $\eta = Z_+X_+Z_-X_-$, and two quadrilaterals $\zeta = X_+Y_+X_-Y_-$. We denote the diagonal points in these quadrilaterals respectively by O_ξ, O_η, O_ζ . Altogether these are six points which, after the transformation has been carried out, coalesce into a single point. Since the segments X_+X_+ and X_-X_- are bisected at M , the four points $X_+X_-X_+X_-$ are the vertices of a parallelogram with center M . On the sides of this parallelogram lie the four points $O_\eta, O_\zeta, O_\eta, O_\zeta$; if we divide the sides at them, we obtain eight segments, equal in fours. Indeed, the four segments issuing from the two points X_+ must be equal, since in the double heptahedron they all come to coincide with one another; and for the same reason the equality of the four segments issuing from the two points X_- follows. Hence $X_+X_-X_+X_-$ is a rhombus and $O_\eta O_\zeta O_\eta O_\zeta$ is a rectangle; the rhombus is circumscribed about the rectangle, and its vertices lie in the angle bisectors of the rectangle diagonals. The same conclusions hold for the quadrilaterals $Y_+Y_-Y_+Y_-$ and $O_\zeta O_\xi O_\zeta O_\xi$, $Z_+Z_-Z_+Z_-$ and $O_\xi O_\eta O_\xi O_\eta$, from which it also follows that the points O_ξ, O_η, O_ζ must be at equal distance from the center M . To the conditions previously derived for the convex cuboctahedron from the assumed existence of a facewise congruent heptahedron there are now added certain conditions in the form of inequalities, obtained from the consideration of the angles formed in the quadrilaterals ξ, η, ζ by the diagonals. Put

$$\angle Y_+O_\xi Z_+ = x, \quad \angle Z_+O_\eta X_+ = y, \quad \angle X_+O_\zeta Y_+ = z, \quad (5)$$

Then, if a facewise congruent heptahedron exists, these angles occur in it as the sides of a trihedral angle; hence they are subject to the inequalities

$$\begin{cases} x + y + z < 2\pi, \\ y + z > x, \\ z + x > y, \\ x + y > z. \end{cases} \quad (6)$$

Conversely, if these inequalities and the preceding equality conditions are satisfied, one may first construct a trihedral angle with sides x, y, z . Then place on its edges and on their backward extensions the pieces occurring in the cuboctahedron quadrilaterals as parts of the diagonals,

$$O_\eta X_+ = O_\zeta X_+, \quad O_\zeta Y_+ = O_\xi Y_+, \quad O_\xi Z_+ = O_\eta Z_+,$$

and on the backward extensions of the edges the segments

$$O_\eta X_- = O_\zeta X_-, \quad O_\zeta Y_- = O_\xi Y_-, \quad O_\xi Z_- = O_\eta Z_-.$$

If the endpoints so laid off are again denoted by $X_+, Y_+, Z_+, X_-, Y_-, Z_-$, it follows at once that the one-sided heptahedron constructed from these points as vertices, with faces denoted by (1), is facewise congruent to the cuboctahedron.

We now carry out the construction of a cuboctahedron of the required kind as follows. Around a point M describe a sphere of radius r , and draw three diameters $O_\xi O_\xi, O_\eta O_\eta, O_\zeta O_\zeta$ not lying in one plane. Denote by ξ', η', ζ' the planes of the rectangles $O_\eta O_\zeta O_\eta O_\zeta, O_\zeta O_\xi O_\zeta O_\xi, O_\xi O_\eta O_\xi O_\eta$, or also these rectangles themselves; and by $x_+, x_-, y_+, y_-, z_+, z_-$ the bisectors of the angles which the diameters lying in ξ', η', ζ' form with one another. These six lines are known to lie, three at a time, in four planes. In choosing their notation one must ensure that the triple $x_- y_- z_-$ lies in one plane; then the same holds for the triples $x_- y_+ z_+, x_+ y_- z_+, x_+ y_+ z_-$. We regard the figure constructed so far as fixed, but now circumscribe about the fixed rectangles ξ', η', ζ' movable rhombi $X_+X_-X_+X_-, Y_+Y_-Y_+Y_-, Z_+Z_-Z_+Z_-$ whose vertices $X_+, X_-, Y_+, Y_-, Z_+, Z_-$ respectively slide on the lines $x_+, x_-, y_+, y_-, z_+, z_-$. As the initial position of such a rhombus we take the one in which its sides are parallel to the diagonals of the inscribed rectangle, hence also equal to $2r$, and are bisected by the vertices of the rectangle. If we fix the three rhombi in

any one of their positions, their vertices form the vertices of a cuboctahedron whose faces are given by (1). In order that this cuboctahedron may be transformable into the double surface of a heptahedron, it remains only necessary that the angles defined in (5) satisfy the inequalities (6).

If, to begin with, we take all three rhombi in their initial position, one sees at once that the cuboctahedron has that special form which can be derived from a parallelepiped with equal edges by cutting off the corners as far as the edge-midpoints. One also sees immediately that it is facewise congruent to the one-sided heptahedron having the six points $O_\xi, O_\xi, O_\eta, O_\eta, O_\zeta, O_\zeta$ as vertices. In the initial position of the rhombi the cuboctahedron is therefore certainly convex and the inequalities (6) hold. If the three rhombi are now allowed to move independently of one another, then certainly, as long as the deviations from the initial position remain within certain bounds, both the convexity of the cuboctahedron and the inequalities (6) will persist. But as long as these inequalities hold, the cuboctahedron can also be transformed into the double surface of a one-sided heptahedron. The most general cuboctahedron of the required kind therefore depends on 7 constants.

Among the different positions which the movable rhombi can assume, besides the initial position the following is also noteworthy: the one in which the sides of the rhombi stand perpendicular to the diagonals of the rectangles, so that the quadrilateral faces of the cuboctahedron are touched by the sphere of radius r . Then the angles x, y, z become equal to the angles between the faces of the corner formed from the edges $MO_\xi, MO_\eta, MO_\zeta$, and the inequalities (6) become inequalities between these angles. It is moreover not difficult to prove that, in the case here considered, these inequalities simultaneously express the conditions for the convexity of the cuboctahedron.

6. Relations to elliptic geometry.

In the preceding discussion we considered two one-to-one correspondences: one between the cuboctahedron and the double surface of the one-sided heptahedron, which is facewise congruent, and one between the simple heptahedron and the plane divided by a complete quadrilateral. The latter correspondence is facewise collinear; it is natural to ask whether in special cases the collineations may become congruences. This can indeed occur, both for ordinary and for elliptic measurement. The latter case is particularly interesting and is closely connected with our last investigations. We shall discuss it more fully in one special case. First we note that the considerations of the preceding paragraph remain essentially unchanged if, instead of ordinary measurement, one takes elliptic or hyperbolic measurement as the basis. In particular, in all three cases the following hold.

1. If through a point O one draws three mutually perpendicular lines and describes about O a sphere of radius r , then the sphere cuts the lines in 6 points, which form the vertices of a one-sided heptahedron composed of regular triangles and quadrilaterals.

2. If through a point M one draws three mutually perpendicular lines, and also the six lines which bisect the angles formed by the first three, then these angle bisectors are cut by a sphere about M of radius ρ in 12 points, which form the vertices of a convex cuboctahedron composed of regular triangles and quadrilaterals. The 24 edges of the cuboctahedron form four regular hexagons whose planes pass through M .

3. Once the radius ρ has been chosen, the radius r can be chosen so that the cuboctahedron and the double surface of the heptahedron become facewise congruent.

Now assume elliptic measurement, and let π denote the length of the whole line. Let the radius ρ then increase, and with it at the same time the radius r , in such a way that the corresponding polyhedra – cuboctahedron and double heptahedron – always remain facewise congruent. At the instant when ρ reaches the length $\frac{\pi}{2}$, every point of the sphere of radius ρ collides with

the opposite point, and the sphere degenerates into a double plane. The cuboctahedron also degenerates into the same double plane, the four regular hexagons becoming double lines. Two coincident points of the double heptahedron again correspond to two coincident points of the double plane, so that we now also obtain a one-to-one correspondence between the simple heptahedron and the simple plane which is facewise congruent. The radius r of the sphere circumscribed about the heptahedron is equal to half the diagonal in the squares and hence, as is easily seen, equal to $\frac{\pi}{4}$.

If the heptahedron is now cut open, as in §1, along the edge path $X_+Y_-Z_+X_-Y_+Z_-X_+$ and spread out in a plane, then the cut places fit together again and the total surface becomes a plane. Henceforth we shall call this heptahedron, and all heptahedra sharing this property with it, *developable*. It should be noted, however, that a continuous transformation of a heptahedron into a plane without cutting is never possible, even if changes of length are allowed during the motion. This rests on the fact that already among closed lines of projective space there exist two essentially different kinds: lines of the first kind (even circuits) are cut by every plane, and indeed by every closed surface, in an even number of points; lines of the second kind (odd circuits) are cut by every plane in an odd number of points. Only lines of the same kind can be continuously transformed into one another without cutting, and likewise a collineation cannot change the kind of a line. Correspondingly one must distinguish surfaces of the first kind, which contain only closed lines of the first kind, and surfaces of the second kind, which also contain closed lines of the second kind. Every closed line or surface lying entirely in the finite part under ordinary measurement, and therefore every line or surface collinear with such a one, is of the first kind. Thus between the heptahedron and the plane there is the purely topological difference that the heptahedron is a surface of the first kind whereas the plane is one of the second kind.

The task with which we shall be concerned here is *the determination of all developable one-sided heptahedra, or equivalently of all region-divisions of a plane which can arise by developing a heptahedron*. In view of the discussion in §4 and of the fact that congruence is only a special case of collineation, it is clear that only such region-divisions can come into question as are produced by a complete quadrilateral.

We first treat the problem for ordinary geometry, retain the previous notation, and assume that a heptahedron and a facewise congruent division of a plane ε are given. To the diagonal points O_ξ, O_η, O_ζ lying in the faces ξ, η, ζ of ε correspond on the heptahedron three points coinciding in a single point O . Because of the congruence of the mapping, according as O is a finite or an infinitely distant point, the points O_ξ, O_η, O_ζ must either all three lie in the finite part or all three be infinitely distant. The latter is excluded, since O_ξ, O_η, O_ζ cannot lie on a single straight line; hence O_ξ, O_η, O_ζ are finite points. On the sides of the diagonal triangle $O_\xi O_\eta O_\zeta$ the vertex pairs $X_+, X_-; Y_+, Y_-; Z_+, Z_-$ of the complete quadrilateral lie so that they are harmonically separated by the vertices of the triangle. Suppose that the quadrilateral vertices lying on the finite sides of $O_\xi O_\eta O_\zeta$ are X_+, Y_+, Z_+ . Then the equations

$$O_\eta X_+ = O_\zeta X_+, \quad O_\zeta Y_+ = O_\xi Y_+, \quad O_\xi Z_+ = O_\eta Z_+$$

must hold, because these pairs of segments lying in ε must be respectively equal to the segments OX_+, OY_+, OZ_+ of the heptahedron figure. Therefore in ε the points X_+, Y_+, Z_+ are side midpoints of the triangle $O_\xi O_\eta O_\zeta$, from which it further follows that X_-, Y_-, Z_- fall to infinity.

We can also derive this result from a general consideration. Suppose there is some polygonal division of the plane ε , and that there exists a polyhedron of the first kind, that is, one containing only even circuits, which is facewise congruent to the division of ε . Then it is easy to see that the line at infinity of ε must be composed entirely of edges of the polygonal division. Otherwise one could draw a straight line g in ε whose point at infinity lay on no edge of the division. This line would be divided by the division of ε into a number of segments, and on the polyhedron there would correspond to it a closed circuit composed of the same number of segments. Like g , this

circuit would contain a single point at infinity, and this point would not be a vertex but would lie in the interior of a segment. Since it cuts the plane at infinity in one point, the closed circuit would be odd, contrary to the assumption made about the polyhedron. This consideration, which can readily be generalized in various directions, shows at once in the present case that one of the lines of the complete quadrilateral must be the line at infinity; from this it again follows that the vertices formed by the other three lines must be the side midpoints of the diagonal triangle $O_\xi O_\eta O_\zeta$.

If we next compare the plane figure with the heptahedron, we see that the angles of the finite triangle $O_\xi O_\eta O_\zeta$ in ε appear in the heptahedron figure as the sides of a trihedral angle. They are therefore subject to the condition that the sum of any two of them be greater than the third; in other words, they must be acute. Since these angles agree with those of the triangle $X_+ Y_+ Z_+$, we have the theorem:

In order that, under ordinary measurement, a division by a complete quadrilateral be able to represent the development of a heptahedron, it is necessary that one line of the complete quadrilateral be the line at infinity and that the finite triangle formed by the other three be acute-angled.

But these conditions are also sufficient. For if they are satisfied, one may first construct a trihedral angle whose sides are the angles of the triangle $X_+ Y_+ Z_+$. If the edges of the corner are then made correspondingly equal to the sides of the triangle, and their endpoints are denoted by X_+, Y_+, Z_+ , while the edges are extended backward beyond the vertex O to infinity and their infinitely distant points are denoted by X_-, Y_-, Z_- , then one immediately sees that the faces of the heptahedron constructed from the last-named points as vertices are congruent to the equally named faces of the division by the complete quadrilateral.

The investigation is quite similar if, instead of ordinary measurement, one uses elliptic measurement. Suppose again that the plane development of a heptahedron is given. Then the comparison of the two figures shows that the following equations must hold among the segments occurring in it:

$$\begin{cases} O_\eta X_+ = O_\zeta X_+, & O_\zeta Y_+ = O_\xi Y_+, & O_\xi Z_+ = O_\eta Z_+, \\ O_\eta X_- = O_\zeta X_-, & O_\zeta Y_- = O_\xi Y_-, & O_\xi Z_- = O_\eta Z_-. \end{cases} \quad (7)$$

Moreover the angles x, y, z specified in (5), since they occur in the heptahedron as the sides of a trihedral angle, must satisfy the inequalities (6). Conversely, if these conditions are satisfied, one may first construct a trihedral angle with sides x, y, z . If one then lays off, on its edges and on their backward extensions, the segments denoted in (7) in the corresponding way, one obtains the vertices of a heptahedron facewise congruent to the division by the complete quadrilateral.

We can give the conditions (6) a more symmetric form by introducing the supplementary angles x', y', z' of x, y, z . They then read

$$\begin{cases} x + y + z < 2\pi, \\ x + y' + z' < 2\pi, \\ x' + y + z' < 2\pi, \\ x' + y' + z < 2\pi. \end{cases} \quad (6a)$$

They say that in each of the four triangles into which the plane is divided by the three lines $O_\eta O_\zeta$, $O_\zeta O_\xi$, $O_\xi O_\eta$, the sum of the angles is $< 2\pi$, that is, the spherical excess is $< \pi$. Since the area of the whole plane is 2π , and the area of a triangle equals its spherical excess, these conditions say that each of the four triangles must have area smaller than half the plane. Thus, if O_ξ, O_η, O_ζ are chosen arbitrarily except for these inequalities, the side midpoints of the corresponding four triangles form the vertices of a complete quadrilateral which represents the development figure of a heptahedron; in this way one has constructed a complete quadrilateral of the required kind in full generality. If, in particular, the vertices O_ξ, O_η, O_ζ are taken at distances $\frac{\pi}{2}$ from one

another, the corresponding triangles become regular and congruent, their angles right angles, and the complete quadrilateral constructed from their side midpoints as vertices gives a division into regular triangles and quadrilaterals. Thus we have the special case that we derived at the beginning of this paragraph as a limiting case of a cuboctahedron. It is also easy to see that the general solution of the problem treated here can be derived in the same way as a limiting case of a cuboctahedron facewise congruent to the double surface of a heptahedron, most simply by starting from the cuboctahedra mentioned at the very end of §5.

We shall also give another solution of our problem, one in which the side conditions troublesome for the construction are avoided. In §4 it was shown that there is only one way of mapping the one-sided heptahedron facewise collinearly onto the division of a plane ε by a complete quadrilateral, and that, when the 7 collineations are continued beyond the bounded faces into the unbounded planes, the 7 complete quadrilaterals given in (4) pass into the complete quadrilateral in ε . Here we are concerned with the case in which these collineations, so far as they refer to the bounded heptahedron faces, are congruences. Then they are also congruences in their full extent, and we have the theorem:

If the one-sided heptahedron is developable, then the complete quadrilaterals given in (4), which the negative octahedral planes $\alpha', \beta', \gamma', \delta'$ cut out from the heptahedron planes, are congruent to the complete quadrilateral obtained in the development.

From the congruence follows the equality of those edges occurring in the complete quadrilaterals which, after suppressing the index ∞ , have the same notation, for example

$$Y_+Z_+ = Y_+^\infty Z_+ = Y_+Z_+^\infty = Y_+^\infty Z_+^\infty, \quad \text{and so on.}$$

It follows again that also the complete quadrilateral cut out on ε^∞ by $\alpha', \beta', \gamma', \delta'$, namely

$$X_-^\infty Y_+^\infty Z_+^\infty, X_+^\infty Y_-^\infty Z_+^\infty, X_+^\infty Y_+^\infty Z_-^\infty, X_-^\infty Y_-^\infty Z_-^\infty,$$

is congruent to the others just named. Now consider the complete quadrilaterals lying in the planes $\alpha', \beta', \gamma', \delta'$:

$$\left\{ \begin{array}{llll} (\alpha') & X_-^\infty Y_+ Z_+, & X_- Y_+^\infty Z_+, & X_- Y_+ Z_+^\infty, & X_-^\infty Y_+^\infty Z_+^\infty; \\ (\beta') & X_+^\infty Y_- Z_+, & X_+ Y_-^\infty Z_+, & X_+ Y_- Z_+^\infty, & X_+^\infty Y_-^\infty Z_+^\infty; \\ (\gamma') & X_+^\infty Y_+ Z_-, & X_+ Y_+^\infty Z_-, & X_+ Y_+ Z_-^\infty, & X_+^\infty Y_+^\infty Z_-^\infty; \\ (\delta') & X_-^\infty Y_- Z_-, & X_- Y_-^\infty Z_-, & X_- Y_- Z_-^\infty, & X_-^\infty Y_-^\infty Z_-^\infty. \end{array} \right. \quad (8)$$

The diagonals occurring in these,

$$X_-X_-^\infty, X_+X_+^\infty, Y_-Y_-^\infty, Y_+Y_+^\infty, Z_-Z_-^\infty, Z_+Z_+^\infty,$$

are identical with the intersection lines

$$(\alpha', \delta'), \quad (\beta', \gamma'), \quad (\beta', \delta'), \quad (\gamma', \alpha'), \quad (\gamma', \delta'), \quad (\alpha', \beta').$$

The diagonal points are therefore the vertices of the tetrahedron $\alpha'\beta'\gamma'\delta'$, and so the point-pairs $X_-X_-^\infty, X_+X_+^\infty, Y_-Y_-^\infty, Y_+Y_+^\infty, Z_-Z_-^\infty, Z_+Z_+^\infty$ are harmonically separated by the vertices of the tetrahedron. This holds for every heptahedron. In the case now under consideration, however, it turns out that in each of the simple quadrilaterals occurring in the complete quadrilaterals (8), for example $X_+^\infty Y_- X_+ Y_-^\infty$, two opposite sides are equal; from this, by congruences of triangles, it follows that the diagonals bisect one another. But the point of intersection is, as we saw, a vertex of the tetrahedron $\alpha'\beta'\gamma'\delta'$.

Considering therefore one of the diagonals, say $X_+X_+^\infty$, we see that the two tetrahedron vertices lying on it are the midpoints of the two connecting segments from X_+ to X_+^∞ . The two

connecting segments of the tetrahedron vertices under consideration are therefore equal to each other, hence equal to half the whole line, that is, to $\frac{\pi}{2}$. Since this must hold for every pair of vertices of the tetrahedron $\alpha'\beta'\gamma'\delta'$, we have the theorem:

If the one-sided heptahedron is to be developable, then the vertices of the associated tetrahedron of the negative octahedral faces $\alpha', \beta', \gamma', \delta'$ must have mutual distances $\frac{\pi}{2}$.

We shall show that this condition is also sufficient, so that the developability of the heptahedron depends solely on the nature of the tetrahedron $\alpha'\beta'\gamma'\delta'$. To make this clear, we start from the construction of an arbitrary one-sided heptahedron. This construction can be carried out by taking the planes $\alpha', \beta', \gamma', \delta', \varepsilon^\infty$ associated with the heptahedron as planes in general position, otherwise arbitrarily. As the intersections of the plane ε^∞ with the edges (α', δ') , (β', γ') , (β', δ') , (γ', α') , (γ', δ') , (α', β') of the tetrahedron $\alpha'\beta'\gamma'\delta'$, we then obtain the points $X_-^\infty, X_+^\infty, Y_-^\infty, Y_+^\infty, Z_-^\infty, Z_+^\infty$, and finally the vertices $X_-, X_+, Y_-, Y_+, Z_-, Z_+$ of the heptahedron as those points on the same edges which are harmonically separated from the first-named points by the tetrahedron vertices.

Furthermore the four planes $\alpha', \beta', \gamma', \delta'$ divide space into 8 regions, bounded tetrahedra, and any fifth plane in general position penetrates, as the complete-quadrilateral division cut out on it shows, into 7 of these regions and not into the eighth. So too does the plane ε^∞ ; and one sees easily, for example from the special case in §1, where ε^∞ is the plane at infinity and the heptahedron lies inside the finite tetrahedron $\alpha'\beta'\gamma'\delta'$, that in every case the heptahedron is completely contained in that one of the 8 regions into which the plane ε^∞ does not enter.

If now the distances between the vertices of $\alpha'\beta'\gamma'\delta'$ are equal to $\frac{\pi}{2}$, then the 8 tetrahedra in question are regular and mutually congruent; the angles occurring in their faces are all right angles; the intersection line of two of the planes $\alpha', \beta', \gamma', \delta'$ is perpendicular to the other two; and two points lying on it are harmonically separated by the vertices lying on it precisely when these vertices are the midpoints of the connecting segments of those points. Two such points are therefore mirror images of one another with respect to each of the two tetrahedron planes that do not pass through them, while with respect to the other two each is of course its own mirror image. Applying this to the point-pairs $X_-X_+^\infty, X_+X_-^\infty, Y_-Y_+^\infty, Y_+Y_-^\infty, Z_-Z_+^\infty, Z_+Z_-^\infty$, we see that the complete quadrilaterals contained in $\alpha, \beta, \gamma, \delta, \xi, \eta, \zeta, \varepsilon^\infty$ and cut out by $\alpha', \beta', \gamma', \delta'$ are interchanged with one another by reflection in these four planes. More precisely, each of the four planes $\alpha, \beta, \gamma, \delta$ is the mirror image of each of the four planes $\xi, \eta, \zeta, \varepsilon^\infty$ with respect to one of the four planes $\alpha', \beta', \gamma', \delta'$. From all this it follows that the 8 complete quadrilaterals are congruent to one another, and therefore also the simple quadrilaterals and triangles which have the same notation after suppressing the index ∞ . Thus the heptahedron with faces (1) is also facewise congruent to the division of ε^∞ .

Accordingly we obtain the following general construction. Take four points A', B', C', D' at mutual distances $\frac{\pi}{2}$, and take a plane ε' passing through none of them and otherwise entirely arbitrary. The planes $\alpha' = B'C'D', \beta' = C'D'A', \gamma' = D'A'B', \delta' = A'B'C'$ divide space into 8 tetrahedra. Into one of these – call it T – the plane ε' does not enter. By repeated reflection of ε' in the planes $\alpha', \beta', \gamma', \delta'$ there arise 7 further planes. The parts of these planes lying in T form a one-sided developable heptahedron for which $\varepsilon^\infty = \varepsilon'$; the development gives the complete-quadrilateral division which occurs both in ε' and in the heptahedron planes.

Of course, in each of the 8 tetrahedra formed by $\alpha', \beta', \gamma', \delta'$ we obtain a developable heptahedron by collecting the parts of the plane ε and of its mirror images contained there. All these heptahedra pass into one another under the reflections, and the corresponding planes ε^∞ are the 8 planes from whose parts they are assembled.

In general, the reflections of an arbitrary point in the planes $\alpha', \beta', \gamma', \delta'$ produce altogether 8 points distributed among the 8 tetrahedra formed by these planes. Again denote one of these tetrahedra by T , and map the whole space eight-to-one onto T by assigning to each point that

one of its mirror points which lies in T . Then a straight line, since it is split by its intersections with $\alpha', \beta', \gamma', \delta'$ into four segments, will have in T as image a closed circuit consisting of four segments. It is clear that this circuit describes the path of a light ray moving inside T and reflected at the boundary faces of T . If a plane pencil of light rays is emitted from a point inside T , then the rays, over their entire course, fill a surface which, as the image of a plane of space, represents a one-sided developable heptahedron. Furthermore, if F is any connected surface in space, its image Φ in T is also connected, and the individual parts of Φ are congruent or symmetric to the corresponding parts of F . On the surface Φ , as in T generally, only even line-circuits are possible.

All these considerations apply, with only very slight modifications, also to the case of ordinary measurement. In particular it follows here as well that the developability of the heptahedron depends only on the associated negative octahedral planes. Of these, three must be mutually perpendicular and the fourth must fall to infinity. Thus take three mutually perpendicular planes α', β', γ' , and also a plane ε' which neither passes through their point of intersection nor is parallel to one of their intersection lines. If one collects all those parts of ε' and of the planes arising from ε' by repeated reflections in α', β', γ' which fall in one and the same octant, then one has constructed, in the most general way, a developable one-sided heptahedron.